AN ABELIAN VARIETY OVER A FIELD IS PROJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file groupoids.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Lemma lemma-algebraic-quasi-projective (canonical tag 0BF7). cited at proof lines 1831-1832. A is quasi-projective over k.
- Lemma more-morphisms-lemma-projective (canonical tag 0B45). cited at proof lines 1832-1833. Proper plus quasi-projective gives projective.

Source reference. Original lemma 0BF7: sources/stacks-chow/tags.txt, source lines 14375–14375. Cited at proof lines 1831-1832: A is quasi-projective; load-bearing.

Source reference. Original lemma 0B45: sources/stacks-chow/tags.txt, source lines 13988–13988. Cited at proof lines 1832-1833: quasi-projective plus properness of the abelian variety gives projectivity; load-bearing.

Lemma 0.2. *Let k be a field. Let A be an abelian variety over k. Then A is projective.*

Proof. This follows from Lemma 0BF7 and More on Morphisms, Lemma 0B45. □

Date: stacks-project:groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.